README — Revision_Deflation_CJT_and_Kohn-Sham_definition_C

PROJECT spacetimes: the 16-component Dirac field with mass term and self-interaction on the 4+4 and 7+1 spacetimes whose coordinates x5, x6, x7 deflate exponentially; Kohn–Sham and CJT analyses

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November 13, 2026

1. What this folder is

The folder holds one sub-project of PROJECT spacetimes. Its hypotheses are the author's:

- The three extra coordinates $x5$, $x6$, $x7$ deflate exponentially in the ordinary time $x4$. The metric is diagonal, and its components along them are $g_{tt}^{(4+4)} = -\mathrm{const}_t^2\, e^{-2\,\mathrm{rate}_t\, x4}$ on the 4+4 spacetime (where they are time-like) and $g_{tt}^{(7+1)} = +\mathrm{const}_t^2\, e^{-2\,\mathrm{rate}_t\, x4}$ on the 7+1 spacetime (where they are space-like), for $t = 5, 6, 7$, with the positive rates $\mathrm{rate}_5 = \mathrm{g55}$, $\mathrm{rate}_6 = \mathrm{g66}$, $\mathrm{rate}_7 = \mathrm{g77}$ and positive constants const_t.
- The Lagrangian of the sixteen-component Dirac field contains a kinetic term, a **mass term** and a **self-interaction** that is quartic in the field, with nine couplings $g_0, \ldots, g_8$.
- The consequences are worked out with two methods, the Kohn–Sham ansatz and the effective potential for composite operators of Cornwall, Jackiw and Tomboulis (CJT), each as a check of the other and of the earlier results.
- **On the 4+4 spacetime the self-interaction is defined by its covariant form.** In the sector of the 4+4 model that can be quantized (fields that do not depend on $x5, x6, x7$) the quartic term can be completed in two ways, which this folder calls **form C** and **form Q**. Form C builds the nine invariants with the covariant adjoint $\tilde{\psi} = \psi^\dagger C^-$, as the invariants of the spacetime; form Q builds them with the adjoint $\bar{\psi} = \psi^\dagger \beta$ of the positive kinetic term. The two differ by the matrix $W_{1644} = iC^-\gamma^4$ in every bilinear, that is, by the sign of the interaction between particles of the two halves of the field. By the author's decision of 13 November 2026, **form C is the definition** in this folder, and form Q is recorded completely beside it.

This folder is the successor of the folder `Revision_Deflation_CJT_and_Kohn-Sham` of the same repository, in which form Q is the definition and form C the alternative; that folder was not changed. Everything in the present folder was computed for it: every number in a document comes from a data file that a command of the program of this folder wrote, and every table is inserted by a script.

What form C changes, in one paragraph (each statement with its numbers is in the documents named in Section 2). Matter polarizes: the self-bound state fills one half of the field and needs two to three times the strength of the couplings that form Q needs. With the filled states of negative energy a field without mass term acquires masses of opposite sign in the two halves. A scalar contact attraction lowers the gap of the fermion pair along the hidden axis and a vector repulsion raises it, the reverse of form Q, so that the self-interaction selects the direction of the order parameter. Every self-interaction of form C breaks the internal symmetry $SO(4) \times U(1)_E$ of fields in ordinary space down to the rotations of $x5, x6, x7$. What does not involve the contact self-interaction (the geometry, the solutions mode by mode, the exchange of the Vorton-Scalar, the cosmology) is the same in both forms, and the 7+1 spacetime has no such choice.

2. Where to start reading

if you want	read
the physics of one spacetime, complete	the manuscript for that spacetime, in `manuscript/`; it defines every symbol it uses and contains its own appendices
the two forms of the self-interaction on 4+4: both written out completely, their exact relation, every consequence in numbers for both, and the discussion of the two	`provenance/P3_two_forms_of_the_self_interaction`; the same in the 4+4 manuscript (Secs. IV, VII, IX, XIII, XV and XVIII D) and in `dirac matrices deflating` (Parts 5 and 6)
the definitions: metric, Dirac matrices printed in full, Lagrangians, field equations, derived step by step for both spacetimes side by side	`dirac matrices deflating`
the meaning of a symbol	`provenance/P1_master_symbols_spacetimes` (every symbol of the project, with one meaning)
how a number was computed, or how to compute it again	`provenance/P2_calculations_code_and_checks`
what was asked, what was done, what was not, and which decisions are open	`provenance/P0_task_rules_deviations_decisions`

Each document exists as `.md`, `.tex` and `.pdf`, and each is complete in itself.

3. Contents

path	content
`manuscript/16 Component Dirac Fermion Condensate Singlet Superfluid on 4+4 Deflating Spacetime`	manuscript for the 4+4 spacetime (`.tex`, `.pdf`, `.md`)
`manuscript/16 Component Dirac Fermion Condensate Singlet Superfluid on 7+1 Deflating Spacetime`	manuscript for the 7+1 spacetime
`manuscript/deflation_refs.bib`	the references of the manuscripts
`dirac matrices deflating`	geometry, Dirac matrices, Lagrangians, field equations, solutions; reports of the checks
`provenance/P0_task_rules_deviations_decisions`	the task, the rules, deviations, decisions, error log
`provenance/P1_master_symbols_spacetimes`	the MASTER registry of symbols and the proof that no symbol has two meanings
`provenance/P2_calculations_code_and_checks`	commands, instructions, data files, figures, complete output of all checks, source files
`provenance/P3_two_forms_of_the_self_interaction`	form C (the definition) and form Q (the alternative): definitions, exact relation, matter, vacuum, pairs, symmetries, gravity; what has been established and what has not; discussion
`provenance/symbols_master.csv`	the registry as a table (246 symbols)
`provenance/generated/`	tables and numbers written by the scripts from the data files
`code/deflation-cjt-ks-c-rs/`	the Rust program `deflation_cjt_ks_c` (36 source files, 17774 lines, 77 unit tests)
`code/scripts/`	independent verification, figures, and the scripts that build and check the documents
`data/`	126 data files and the reports of the checks
`figures/`	38 figures (`.pdf` and `.png`)
`docs_src/`	the sources of all documents

4. How to run everything

What is needed. A Rust toolchain (`cargo`); Python 3 with numpy, scipy, sympy, mpmath and matplotlib; for the documents, pandoc, XeLaTeX with latexmk, and pdflatex with REVTeX 4.2 and BibTeX. The program uses the Rust crate `code/spacetimes-rs` at the top of the repository as a library; nothing is downloaded.

Where. All commands are run from this folder. On the author's computer:

```
cd "/Users/nsh/Developer/github/spacetimes/Revision_Deflation_CJT_and_Kohn-Sham_definition_C"
```

Step 1. Compile the program and run its unit tests.

```
cargo test  --release --manifest-path code/deflation-cjt-ks-c-rs/Cargo.toml
cargo build --release --manifest-path code/deflation-cjt-ks-c-rs/Cargo.toml
```

The first command must end with `test result: ok`. The second builds the program `code/deflation-cjt-ks-c-rs/target/release/deflation_cjt_ks_c`.

Step 2. Run the commands of the program. Each writes its data files and its report of checks into `data` and ends with a line such as `kohnsham: 13 of 13 checks passed`.

```
B=code/deflation-cjt-ks-c-rs/target/release/deflation_cjt_ks_c
$B help
for c in selfcoupling kohnsham pairing gravity matrices modes symmetry channels \
         history quench paired cosmology matter background spectrum; do $B $c; done
$B forms
$B forecast
$B superfluid
$B families
```

The first line abbreviates the path of the program, and `help` lists the commands with a description of each. The loop runs fifteen commands, each of which takes from a few seconds to about two minutes. The last four take longer (about ten minutes, a quarter of an hour, three quarters of an hour, and an hour and a quarter on a machine with two processors). The command `forms` computes the two forms of the 4+4 model side by side.

Step 3. The independent verifications.

```
python3 code/scripts/verify_deflating_independent.py
python3 code/scripts/verify_cjt_ks_independent.py
python3 code/scripts/mutation_test_independent.py
python3 code/scripts/vendor_eb.py --check
```

The first two repeat the central results with computer algebra and general-purpose numerical tools (43 of 43 and 45 of 45 checks pass). The third introduces deliberate errors into copies of the two scripts, one at a time, and records whether the checks notice (45 of 45 detected; about a quarter of an hour). The fourth verifies that four modules of the program are exact copies of modules of the library crate with a listed set of changes.

Step 4. Figures, tables, documents, and the checks of the documents.

```
python3 code/scripts/make_figures.py
python3 code/scripts/write_symbol_registry.py
python3 code/scripts/write_doc_tables.py
python3 code/scripts/assemble_docs.py
python3 code/scripts/build_manuscripts.py
bash    code/scripts/build_docs.sh
bash    code/scripts/check_all.sh
```

In this order: draw every figure from the data files; write the registry of symbols; write every table and every computed number of the documents from the data files; assemble the Markdown documents; assemble and typeset the two manuscripts; typeset the other documents; run the checks of the documents and read the verdict of every report. The last command prints one line for each check and must end with `ALL PASS`. The single command `bash code/scripts/build_all_docs.sh`

runs all of the steps after the figures in the right order, twice where a report that a document prints is itself made from the documents, and ends with the same verdict.

5. The checks, in numbers

what	result
checks of the program (exact where possible; reports `data/*_checks.txt`)	310 of 310 pass
unit tests of the program	77 pass
independent verification, geometry, Dirac field, modes, cosmology	43 of 43 pass
independent verification, self-interaction, Kohn–Sham, CJT, pairing, gravity	45 of 45 pass
deliberate errors introduced into the independent scripts	45 of 45 detected
checks of the registry of symbols, of the documents and of the data files against it	8 of 8 pass
numbers of the running text traced to a statement of a program	660 numbers in 29 documents, 0 not traced

6. Conventions in one place

- Coordinates $x0, \dots, x7$: $x1, x2, x3$ ordinary space, $x4$ ordinary time, $x0$ a hidden space-like coordinate, $x5, x6, x7$ the deflating coordinates.
- Sign of the metric: space-like components positive, time-like components negative.
- The rates g55, g66, g77 are written upright and never with a subscript; a component of the metric carries the superscript $(4+4)$ or $(7+1)$; a coupling of the self-interaction, $g_0, \dots, g_8$, carries one digit as its subscript.
- Units: $\hbar = c = 1$.
- On the 4+4 spacetime: **form C** is the self-interaction built with the covariant adjoint $\tilde{\psi} = \psi^\dagger C^-$ (the definition; tag 44c in checks and data files), **form Q** the one built with $\bar{\psi} = \psi^\dagger \beta$ (the alternative; tag 44q); the **halves** of the field are the eigenspaces of the matrix W_{1644} with the eigenvalues $+1$ and -1; **set A** is the scalar coupling $g_0 = -g$ alone, **set B** is $g_0 = -g$ with $g_1 = +0.1\, g$, with a strength $g > 0$.
- In the output of the programs, “the main project” means the part of PROJECT spacetimes that treats the flat 4+4 and 7+1 spacetimes; its program is used as a library, and several checks compare with its data files.

7. What is open

The form of the self-interaction on the 4+4 spacetime is decided: form C. Which of the two forms is the right one for the model is not settled by any calculation; the considerations on both sides are set out in the documents named in Section 2. Not computed for form C: states with coherence between the two halves of the field, the potential of the order parameter with contact couplings, and pairing inside polarized matter. Matters that need the author’s decision (the role of a bare mass beside the Vorton-Scalar, the reading of the couplings of the reference example, and what is to be done about a finding that concerns the earlier folders: their gap equation was solved with an upper limit of the momentum integral that is too low) are set out, with the deviations from the instructions and the log of errors, in the document P0.